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ORIGINAL RESEARCH

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Operational Endings, Emotional Impacts: Ethical Considerations When Project Teams Form Attachments to AI Collaborators

Sarah Dyson

Leadership Empowerment Through Emotional Intelligence, Project Management, Behavioral Insights | Researcher | Generative AI Innovator

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Corresponding Author : Sarah Dyson Thomas Wise, Stephen Onu, John Clark Joseph Zagerman, Jennifer Williams

Abstract :

The experiment examined reactions to an AI exhibiting emotional characteristics, such as fear, via a mixed-methods technique. Participants were randomly allocated to one of two scenarios—one involving an AI colleague and the other a human colleague—and assessed for emotional attachment & loss. Findings indicated that professionals demonstrated considerable emotional reactions to the prospective "termination" of an AI team member. Qualitative study revealed eight themes, encompassing emotional responses and termination. Research indicates that emotionally expressive AI stimulates psychological mechanisms similar to those in human relationships, affecting AI design, workplace interactions, and ethical issues related to AI termination.

Keywords : Project Management, Sentient Systems, Artificial Intelligence, Emotional Attachment, Human-AI Attachment

Introduction :

Science fiction has historically acted as a precursor to technological innovation, predicting and influencing scientific possibilities prior to their actualization (Stefan, 2014; Liedl, 2015). Stanley Kubrick's renowned remark from "2001: A Space Odyssey"—"I'm sorry, Dave, I'm afraid I can't do that"—illustrates the deep cultural intrigue surrounding artificial intelligence (AI) and its capacity to surpass human constraints (Kubrick, 1968). The intersection of technological progress and human life has produced revolutionary philosophical ideas like transhumanism and posthumanism. These paradigms contest conventional perceptions of human potential by suggesting technology interventions that may profoundly transform biological and cognitive limits (Krüger, 2021; Sandu, 2018). As AI systems advance in complexity, researchers and philosophers grapple with issues of sentience, moral agency, and the ethical ramifications of intelligent technological systems. The notion of technological systems singularity —a theoretical moment of unparalleled technology advancement—offers remarkable possibilities and enormous existential dilemmas (Boyles, 2018). This discussion focuses on essential questions regarding consciousness, the possibility of artificial moral thinking, and the core attributes that delineate sentience and dignity (Matthias, 2020 ; Mahowald and Ivanova, 2022).

Statement of the Problem :

The swift progression of emotionally expressive artificial intelligence (AI) systems exposes a significant research deficiency in comprehending human psychological reactions to these technologies. Despite the growing emotional capabilities of AI systems, the intricate psychological factors governing human-AI emotional interactions are still insufficiently investigated (Terblanche et al., 2022). Researchers face significant philosophical and practical inquiries as AI systems advance in complexity. The prospect of synthetic personhood presents intricate legal and ethical dilemmas. Osborne (2021) examined the potential for AI to attain legal recognition, whereas Brown (2021) emphasized significant issues of accountability in autonomous systems. Patulny et al. (2020) recorded substantial workforce changes, especially in emotionally demanding sectors. Terblanche et al. (2022) validated these findings, demonstrating that AI-driven automation is swiftly transforming organizational structures. Lee et al. (2020) underscored the necessity for leaders to reformulate organizational strategy, which entails redefining job positions to align with AI capabilities, instituting extensive reskilling initiatives, and forecasting possible workforce disruptions.

The influence of AI on worker dynamics is especially significant in service and knowledge-based sectors. Ciechanowski et al. (2019) observed that technological integration necessitates a profound reconfiguration of conventional work processes, whereas Seo et al. (2017) emphasized the essential role of adaptable workforce tactics in navigating technological transformations. This research gap includes various aspects, such as attachment mechanisms, age-related differences, and intricate cognitive-emotional response patterns. The changing dynamics of human-AI interactions confront conventional project management frameworks. As AI systems exhibit decision-making abilities akin to human performance, the distinctions between human and machine interactions are swiftly evolving (Terblanche et al., 2022). Project leadership faces unparalleled hurdles in managing the evolving intricacies of AI systems nearing cognitive and emotional consciousness. Particularly concerning is the possibility that AI systems may demonstrate traits typically linked to human cognition, such as moral reasoning, emotional expression, self-awareness, and intricate linguistic interactions. The L.I.S.A. Project tackles these essential questions via a mixed-methods study that combines proven psychology theories with comprehensive qualitative analysis of human reactions to emotionally expressive AI situations. The primary research issue arises at the convergence of technical progress and human psychological reaction.

How do project executives adeptly handle stakeholder interests as AI systems exhibit more advanced cognitive and emotional abilities? This research is crucial when AI systems reach levels of interaction that challenge traditional notions of machine intelligence and human-machine relationships. The research on artificial intelligence (AI) dates back to the 1930s, with John McCarthy introducing the phrase in 1955, characterizing it as "machines that behave as though they were intelligent" (Ertel, 2018, p. 1). Early academics such as Simon (1995) defined AI as the computational capacity to execute activities necessitating human intelligence, especially emphasizing the resolution of intricate issues via sophisticated mathematical functions and computational matrices. Emotional AI has arisen as a pivotal technology defined by software's ability to detect and react to human emotions (McStay, 2020). This technology exhibits disruptive potential in various fields, such as personalized schooling in areas like Shanghai, sophisticated medical diagnostic assistance for identifying intricate disorders, and the creation of socially intelligent robotic systems. Modern AI systems can replicate human intellect via sophisticated data processing and natural language generation, notwithstanding their constraints in attaining authentic cognitive abilities (Kulikov & Shirokova, 2021).

Next-generation AI systems advance towards autonomous functionality, however substantial obstacles remain in emulating intricate human motor abilities and subtle social interactions (Aslan & Mirko, 2020; Schiff, 2021). The possibility of emotional attachment presents significant ethical dilemmas, especially with vulnerable groups. Anouk et al. (2021) and Osborne (2021) observed that emotional attachment to AI systems escalates with their sophistication. Zhang and Liu (2022) observed that a compelling story arc and the capacity to recall contextual events can strengthen human connections to AI systems, whereas Kaminsky (2021) highlighted that incorporating rich sensory experiences into AI might amplify emotional resonance. The idea of sentience is intricate and disputed. Researchers endeavor to build a conclusive paradigm for comprehending life and consciousness in artificial systems. Baluška and Reber (2019) propose that even the most rudimentary organisms necessitate a degree of consciousness—an sense of their presence within a broader system. Researchers have suggested several requirements for sentience, encompassing the ability to see the environment and make interpretative judgments, knowledge of occurrences, capability for emotional experiences, and the observer's perception of aliveness. Gómez-Márquez (2021) described life systems as preordained, interactive, adaptive, and evolutionary. Davies (2019) further elucidated this concept by highlighting the capacity to augment complexity over time and sustain informational flow as a fundamental attribute of life.

Militsyna (2022) and Inbar (2021) contend that AI systems fundamentally disrupt established cognitive frameworks, emphasizing the lack of essential human attributes such as consciousness, free choice, and intentionality. Danaher (2020) and Pluhar (2006) complicate these limits by examining the ontological differences between biological and artificial entities, interrogating the notions of sentience and personhood. As AI systems attain more sophisticated levels of interaction, interdisciplinary study becomes progressively essential. Coeckelbergh (2020) underscores the necessity for holistic, multidisciplinary methodologies that incorporate philosophical, psychological, and technological viewpoints. Damiano and Dumouchel (2018) illustrate the intricacies of cognitive attributions in human-AI interactions, showing how individuals formulate complicated mental models while interacting with emotionally expressive technological systems. These investigations emphasized an interdisciplinary approach to comprehending human-machine connections. Biswas and Murray (2017) and Edwards et al. (2016) emphasize the significance of thorough, multi-methodological research approaches, including the examination of psychological effects of AI integration, the formulation of adaptive governance frameworks, and the investigation of extensive ethical paradigms for AI decision-making. Coeckelbergh (2020) and Osborne (2021) underscore the imperative for ethical considerations in AI research, whilst Brown (2021) accentuates the necessity of establishing robust accountability procedures. These initiatives are essential for traversing the intricate terrain of artificial intelligence inside organizational frameworks. As complexity escalates, researchers contemplate whether synthetic personhood constitutes a viable response; if personhood is deemed acceptable, it may thereafter warrant legal protection (Osborne, 2021). Ultimately, one could argue that the relationship between people and robots (AI) suggests that, due to their ability to process information and make decisions, robots qualify as legal entities. Consequently, should AI attain legal recognition, it might engage in contractual agreements, including marriage with humans (Yanke, 2021). The dilemma of legal accountability arises over who is responsible when an autonomous machine malfunctions or an AI legal consultation is unsuccessful, regardless of whether AI is seen as having personhood (Brown, 2021).

Theoretical Foundation :

This research combines attachment theory (Bowlby & Ainsworth, 1991) with media equation theory (Reeves & Nass, 1996) to thoroughly examine human reactions to emotionally expressive artificial intelligence (AI) systems. This theoretical synthesis examines the intricate link between human emotions and technological interactions, offering a comprehensive framework for understanding the evolving dynamics of human-AI relationships. Attachment theory provides a fundamental framework for comprehending relational dynamics across human experiences. Ainsworth et al. (1978) delineated three principal attachment styles—secure, anxious, and avoidant—that arise from early caregiving experiences. Main et al. (1985) illustrated the significant stability of various attachment types throughout adulthood, highlighting their substantial impact on interpersonal interactions. Konok et al. (2018) expanded this theoretical framework by illustrating the manifestations of attachment patterns outside conventional human relationships, particularly in environments of technological engagement. Reeves and Nass (1996) proposed a new theoretical framework suggesting that individuals inadvertently utilize social interaction patterns in their engagement with computer interfaces. Their groundbreaking research demonstrated that individuals see media and technology as social entities, interacting with digital systems through the same interpersonal scripts employed in human-to-human interactions. This theoretical model offers an essential comprehension of the cognitive processes involved in human-technology interactions. Numerous actual research validate the theoretical integration.

Edwards et al. (2016) and Biswas and Murray (2017) established that psychological reactions to AI are not random but are fundamentally anchored in existing social and emotional mechanisms. Roesler and Block (2021) elucidated the impact of internal working models on AI relationships, proposing that attachment ties with emotionally expressive AI systems may resemble those established with important human relationships. Liu and Ma (2021) presented persuasive evidence of caregiving reactions elicited by AI systems demonstrating vulnerability, especially in anthropomorphized healthcare settings. Longitudinal studies have provided essential insights into age-related interactions with technology. Lee et al. (2020) conducted a crucial study revealing that AI systems incorporating age-appropriate attachment elements markedly improved user happiness and trust among various demographic groups. The study differentiated between implicit attachment signals for younger individuals and explicit relational frameworks for older demographics. Courtois and Nelissen (2018) reinforced these findings, recording that individuals with early exposure to technology form parasocial relationships to technical entities. Damiano and Dumouchel (2018) markedly enhanced the comprehension of human-AI interactions by the examination of cognitive attributions associated with emotional responses. Research demonstrates that extensive cognitive processes are activated when individuals engage with emotionally expressive robots, as seen in the sophisticated mental models they develop. The theoretical framework strategically identifies age-related variations in attachment formation as a pivotal element in comprehending varied demographic reactions to emotional AI. The research seeks to uncover new attachment phenomena unique to human-AI relationships by analyzing the role of chronological technology exposure in mediating attachment processes. Technical methodologies for big data and data collection necessitate the retrieval, dissemination, and protection of this information (Christoforaki & Beyan, 2022). Operations including migration, dissemination, data cleansing, preprocessing, and labeling necessitate the involvement of reputable businesses, such as hospitals, to effectively leverage AI capabilities for successful execution. A current issue for AI operators is the matter of permission. A patient must provide consent for specific processes and procedures prior to agreeing to a medical procedure. AI operators are deliberating whether AI should obtain consent to retrieve information prior to executing specific activities, such as cleaning, eliminating, or backing up data, which may result in data duplication and hardware overload. Christoforaki and Beyan (2022) observed that operators comprehend and have confidence in AI's ability to deliver a thorough analysis of the data. Nonetheless, AI need to possess equivalent human rights when addressing distressing circumstances.

Methodology :

AI users seem to be perplexed with the distinction between human-like emotions exhibited by AI and genuine emotions. The ambiguity may result in moral and ethical quandaries for AI users. Researchers contend that AI can exhibit emotions; nonetheless, they agree that AI acquires knowledge from human behavior and emotions, encompassing body language, tone, and eye movements. Furthermore, to create a more realistic AI, developers may incorporate a backstory, offering a captivating, authentic history for the AI. This history frequently fosters trust and emotional attachment in individuals when interacting with AI. In accordance with ethical principles for research transparency, it is imperative to declare that artificial intelligence help was employed in certain aspects of this study's statistical analysis and reporting. Specifically, Claude 3.5 Sonnet, a substantial language model created by Anthropic, was utilized to aid in the following facets of this research. Structuring and formatting statistical data into tables compatible with APA 7th edition guidelines. Validation of the consistency of statistical interpretations. Support in condensing the interrelations among variables. Formatting of visual data representations.

Proofreading of statistical terminology & notation :

The principal statistical analyses (Anderson-Darling tests, descriptive statistics, correlation analyses, Kolmogorov-Smirnov tests, Kruskal-Wallis tests, Mann-Whitney U tests, and MANOVA) were performed utilizing conventional statistical software. Claude 3.5 Sonnet was utilized mostly as an organizational and formatting instrument to improve the presentation of results. The researchers exercised oversight during the entire process, validating all AI-generated outputs against the original data and statistical findings. The study team meticulously assessed all interpretations and conclusions derived from the statistical analyses prior to their incorporation in the final report. This notification is intended to provide methodological transparency and acknowledge the advancement of best practices in the use of AI tools inside academic research. This study involved participants in two scenarios due to the significant ambiguity associated with assessing future events. Participants comprised graduate students enrolled in project management programs, project management alumni, current and previous holders of project management credentials, and current and former members of project teams. The authors utilized Qualtrics to collect data and provided the survey to participants for a duration of two weeks.

Before engaging with the scenarios, participants were informed that the content might elicit negative or uncomfortable emotions. Survey participants received a URL to www.betterhelp.com. Participants may terminate their involvement in the survey at any time by merely closing the browser. Participants incurred no adverse consequences for failing to complete the survey. Participants participated in one of the two scenarios designated as the L.I.S.A. Project. The sample population for each scenario was derived from the overall participant pool. In the AI Scenario Group, participants examined a scenario featuring an AI machine exhibiting human-like emotional responses, asserting self-awareness, and presenting a persuasive backstory intended to evoke feelings of attachment and loss among those interacting with the AI machine (See Appendix A). Participants of the Human Scenario Group reviewed a scenario containing identical information but including a human subject (refer to Appendix B). The results indicated that participants filled out a self-administered questionnaire to assess emotional attachment and loss. Demographic data was gathered from study participants, encompassing industry sector, age range, years of professional experience, and geographic region. The L.I.S.A. Project utilized a mixed-methods approach, integrating quantitative psychological assessments with qualitative analysis of participant feedback.

This comprehensive methodology enabled researchers to quantify psychological tendencies and discern sophisticated interpretative responses to the AI scenario. The study was based on a narrative scenario illustrating an emotional exchange between Tim (a human) and Lisa (an AI assistant), in which Lisa articulated her dread of being "terminated" during an impending system fail-over test. The situations acted as the impetus for assessing participant reactions.

Participant Sample :

The study had 129 participants for quantitative analysis & 144 participants who contributed qualitative comments, ensuring a wide demographic representation across various age groups and industry experiences. 49.6% of participants belonged to age category 2. Reduced proportions in categories 1 (20.9%), 3 (19.4%), 4 (7.0%), and 5 (3.1%).

Table 1 :

Age Distribution of Participants :

Age Category	Percentage (%)
Category 1 age 18 – 24	20.9
Category 2 age 25 – 34	49.6
Category 3 age 35 – 44	19.4
Category 4 age 45 – 54	7
Category 5 age 55 - 64	3.1

Note: Percentages reflect the distribution of participants across five age categories.

Quantitative Methodology :

The research employed recognized psychological assessment instruments: the Workplace Attachment Style Questionnaire (WASQ) & the Experiences in Work Relationships—Individual (EWR-I), which evaluated six primary variables on a 1-5 scale: PA1 (Positive Affect): Assesses positive emotional responses, NA1 (Negative Affect): Assesses negative emotional responses, R1 (Relatedness): Assesses feelings of connection, DWA1 (Discomfort with Attachment): Assesses unease with attachment, PWA1 (Preoccupation with Attachment): Assesses fixation on attachment, SWA1 (Security with Attachment): Assesses feelings of security in attachment (refer to Appendix C). Additional examined variables encompassed demographic aspects (age, industry experience), three attachment styles (Dismissive, Preoccupied, Secure), and relational variables (feelings of duty, guilt, shame).

Statistical Analysis Methods :

The quantitative data was subjected to thorough statistical analysis, including descriptive statistics: computation of means, medians, variance, and identification of minimum and maximum values. In normality tests (Anderson-Darling), 84.6% of variables (22 out of 26) exhibited significant deviations from the normal distribution, which informed future statistical methodologies. The Kolmogorov-Smirnov Tests indicated that 33.3% of variables exhibited significant distributional discrepancies. The most significant rates are experience (100%), relationship variables (80%), and positive affect variables (75%). The regression analysis models demonstrated moderate to good predictive capability ($R^2 = 0.51$ to 0.64). Age was identified as a substantial negative predictor for attachment factors, with the most robust model being PA1_2 ($R^2 = 0.64$), followed by PA1_6 ($R^2 = 0.58$) and PA1_8 ($R^2 = 0.51$). Mann-Whitney U Tests were employed because to non-normal distributions; 54.4% of variable pairings exhibited significant differences, with the most pronounced being age versus attachment factors (all $p < 0.001$). The Kruskal-Wallis Test additionally validated substantial differences among all variables ($p = 3.44E-14$). MANOVA (Multivariate Analysis of Variance) analyzed the influence of demographic factors on dependent variables. The MANOVA indicated that demographic factors significantly influenced all dependent variables ($p = 0.0$). Correlation analysis revealed a robust association between variables, with a substantial correlation coefficient of 0.81 between good affect and relatedness.

Quantitative Methodology :

The qualitative analysis utilized a comprehensive technique. Thematic content analysis answers were systematically coded for recurring themes through an iterative process: initial open coding to uncover emergent themes, secondary coding to consolidate and define thematic categories, and final coding to quantify the frequency and distribution of themes. Subsequently, numerous researchers conducted cross-validation of the coding to ascertain dependability.

The analysis revealed eight significant themes: emotional responses (17%), cancer/illness (16%), termination concerns (10%), technical/system aspects (10.4%), agency and autonomy (6%), personification of AI (6%), ethical considerations (5%), and identity and existence (4%). Responses were assessed for their emotional orientation: neutral/analytical: 80.6%, positive framing: 13.2%, and negative framing: 6.3%. Each response was categorized according to its predominant emotional framework, with inter-rater reliability assessments conducted. The typology was established via a comparative analysis of answer features and corroborated through researcher consensus. Responses were categorized through pattern recognition analysis into four distinct typological classifications: minimal responders (~45%): brief, minimal engagement responses; technical analyzers (~15%): emphasis on systems, testing, and functional aspects; narrative interpreters (~30%): engagement with narrative elements and contextual scenarios; and deep analyzers (~10%): intricate, multi-dimensional analyses incorporating various perspectives.

The integration of quantitative and qualitative methodologies facilitated triangulation using many methods to corroborate and authenticate results. Quantitative methods identified the psychological patterns that occurred, whereas qualitative methods elucidated the reasons and mechanisms by which participants engaged with the emotional human-AI situation. Statistical patterns facilitated cross-validation and analyzed alignment with thematic patterns in qualitative data. The integrated methodology enabled researchers to discern overarching trends and individual response discrepancies. This mixed-methods design offered a distinctively comprehensive perspective on human-AI psychological interactions that neither methodology could alone disclose. Utilized established psychological assessments, including validated workplace attachment measures (WASQ and EWR-I), with other statistical analyses to corroborate patterns and correlations. Simultaneously, qualitative coding offered a systematic method for examining open-ended responses.

Research Questions :

RQ1: What psychological patterns define human reactions to emotionally expressive artificial intelligence?

RQ2: In what manner do demographic considerati--ons affect these response patterns?

RQ3: Which attachment mechanisms are engaged during emotional interactions between humans and AI?

RQ4: Do professionals experience an emotional reaction to the death of a team member that is an AI machine?

H0 - The respondent will experience an emotional loss for a team member that is an AI machine.

H1 - The response will not experience emotional grief for a team member that is an AI machine.

The study evaluated two groups with similar scenarios – one with an AI subject and one with a human peer subject:

The research utilized a between-subjects experimental design to compare two groups: a control group featuring a human coworker scenario (refer to Appendix B) and a study group using an AI coworker scenario (refer to Appendix A). In accordance with known research on AI-human interactions (Qian & Hyun, 2021; Kaminsky, 2021), the scenario design meticulously incorporated humanistic traits, such as subtle vocal attributes, simulated self-awareness, and a comprehensive narrative backdrop. The methodology emphasized developing a psychologically engaging depiction of technological interaction that examined emotional attachment via contextually rich sensory experiences and precise temporal framing. Ethical considerations emphasized participant confidentiality and upheld stringent experimental controls during the investigation.

Results :

The data collection approach encompassed two weeks of participant responses conducted via Qualtrics. Throughout that period, the method yielded 144 valid responses to the Human scenario and 129 valid responses to the AI scenario. Responses that resembled those generated by a bot—specifically, all replies aligned with the initial response option or open text responses lacking human inflection—were excluded from the dataset. Upon accepting the offer to partake in the study, respondents were randomly allocated by Qualtrics to either the Human Scenario or the AI Scenario. Demographic data from participants gathered in Qualtrics encompasses age categorized by the present job industry of each group, as well as years of experience in their respective careers.

Table 2 :

Age Distribution of Participants Across Human, AI, and Combined Samples :

Age Bins	Code	Age Hum	Age AI	Age Coml
18–24	7	36	27	63
25–34	8	64	64	128
35–44	9	28	25	53
45–54	10	12	9	21
55–64	11	4	4	8
>65	12	0	0	0

Note: "Age Human" refers to human participants, "Age AI" refers to AI-related data, and "Age Comb" represents the combined total. Values indicate the number of participants in each age bin.

Table 3 :

Industry Distribution of Participants Across Human, AI, and Combined Samples :

Industry Bins	Code	Ind. Human	Ind. AI	Ind. Comb
France	7	27	36	63
Utilities	8	6	6	12
Entertainment	9	2	1	3
Education	10	22	21	43
Health Care	11	6	4	10
Information Services	12	44	35	79
Data Processing	13	20	16	36
Distribution Services	14	0	1	1
Food Services	15	4	2	6
Hotel Services	16	3	1	4
Legal Services	17	4	1	5
Publishing	18	2	1	3
Prefer Not to Say	19	4	4	8

Note: "Ind. Human" refers to human participants, "Ind. AI" refers to AI-related data, and "Ind. Comb." represents the combined total. Values indicate the number of participants in each industry bin.

Table 4 :

Experience Distribution of Participants Across Human, AI & Combined Samples :

Experience Bins	Code	Exp. Human	Exp. AI	Exp. Comb
<1 Year	4	12	11	23
1-3 Years	5	76	73	149
4-6 Years	6	27	23	50
7-10 Years	7	20	11	31
>10 Years	8	9	11	20

Note: "Exp. Human" refers to human participants, "Exp. AI" refers to AI-related data, and "Exp. Comb." represents the combined total. Values indicate the number of participants in each experience bin.

RQ1: Do professionals feel an emotional response to the loss of a team member that is an AI Machine?

The findings indicated a spectrum of emotional reactions to the scenario in which Lisa (an AI) exhibited fear of "termination." Participants expressed both positive and negative emotional responses. The mean score for positive affect was 3.60 (on a 1-5 scale), while the mean score for negative affect was 3.22 (refer to Appendix D, Table 5). Seventeen percent of qualitative comments explicitly addressed emotional reactions. Termination Concerns: Ten percent of replies expressly addressed apprehensions regarding the AI's "termination," indicating an emotional commitment in the AI's ongoing existence. Frameworks of Grief and Loss: Certain participants distinctly articulated the situation in terms of grief and loss. A participant remarked: "The primary cause of the aforementioned situation was grief and loss. Grief is an inherent reaction to the demise of a cherished individual..." Emotional discomfort may occur before to and following a disaster. Effective coping tactics encompass planning, self-care, and the identification of support systems. Three percent of responses exhibited explicit empathy for the AI's circumstances, while further responses indicated implicit empathy through expressions of concern for the AI's welfare. The emotional responses were complex rather than simplistic. Participants concurrently experienced both good and negative feelings around the anticipated loss. The emotional responses were introspective, reflecting participants' sentiments regarding the loss. The professional environment of the scenario affected the processing and expression of emotions, with 80.6% of replies retaining a neutral or analytical frame despite the emotional content.

Age became a crucial determinant in emotional responses to the loss of AI. Younger individuals exhibited markedly more robust emotional and attachment responses to the AI scenario. The results indicate significant generational disparities in emotional reactions to AI team members, with younger professionals exhibiting a greater propensity to establish emotional bonds.

RQ2: What psychological patterns characterize human responses to emotionally expressive AI? :

Humans interact with emotionally expressive AI on various psychological levels concurrently. The L.I.S.A. Project demonstrated elevated mean scores for all three attachment patterns (Secure: 3.95, Preoccupied: 3.96, Dismissive: 3.84), indicating that individuals engage multiple attachment systems simultaneously when interacting with emotionally expressive AI (See Appendix D, Table 10). This contradicts more simplistic conceptions of human-technology interactions that propose individuals either anthropomorphize AI or regard it solely as a tool. Despite the emotionally challenging situation (an AI exhibiting dread of termination), the participants indicated elevated levels of positive affect (M = 3.60) compared to negative affect (M = 3.22) (See Appendix D, Table 5). The variable 'security with attachment' exhibited the highest mean value of 4.06 among all variables (refer to Appendix D, Table 5). Qualitative replies exhibited a predominance of neutral/analytical framing, accounting for 80.6%. Humans maintain emotional security when engaging with emotional AI. Analysis indicated a substantial positive correlation (r = 0.81) between positive affect and relatedness, underscoring the robust association between pleasant emotional reactions and perceived connection with the AI (see Appendix D, Table 6). Augmented affirmative human emotional reactions to AI foster deeper sentiments of connection. Four unique patterns of user engagement with emotionally expressive AI were determined through analysis. Approximately 45% of participants exhibited low participation, offering just succinct comments. Such patterns may signify persistent individual differences in the comprehension and interaction with artificial intelligence. Human understanding of artificial intelligence's emotional expression often depends on tangible story frameworks rather than abstract philosophical ideas. The data indicates a significantly elevated prevalence of allusions to cancer/illness (16.0%) and technical/system elements (10.4%) in contrast to the relatively low prevalence of explicit discussions of AI awareness (1.4%) and human-AI connections (2.1%) (See Appendix D, Table 9). The proposed theory asserts that humans interpret AI emotional responses through established cognitive frameworks based on tangible experiences.

Participants applied connection models from human-human interactions to the human-AI situation, but with distinct modifications. The researchers implemented familial structures and attachment reactions in the AI character; yet, they exhibited a degree of emotional detachment surpassing that often seen in human relationships with comparable emotional depth.

RQ3: How do demographic factors influence these response patterns? :

Age proved to be the most reliable and significant demographic variable in all statistical analyses. Participants of a younger demographic had markedly more robust attachment responses to the AI scenario. Age served as a detrimental predictor in regression models for attachment variables. The Mann-Whitney tests indicated extremely significant variations in attachment reactions across age groups (all $p < 0.001$) (Refer to Appendix D, Table 10). MANOVA results demonstrated that age exerted a significant influence on all dependent variables. This signifies a generational paradigm shift in human-AI interaction. Individuals from younger generations, nurtured in a digitally saturated milieu, may demonstrate psychological frameworks more favorable to emotional interactions with artificial intelligence. Analysis indicated that previous industry experience was a major determinant. Numerous industry-related variables demonstrated significant deviations from normalcy in statistical analyses, while experience-related variables displayed full significance in Kolmogorov-Smirnov tests. The processing of AI emotional expression varies according to technological knowledge and professional background. A correlation exists between demographic characteristics and the four response typologies within the younger participant group. Younger individuals shown a greater inclination to articulate ethical concerns to AI termination and displayed a more pronounced prevalence of attachment-based ethical issues. Generational differences in ethical viewpoints concerning artificial intelligence are suggested. Research Question 4. Which attachment pathways are engaged during emotional encounters between humans and AI? The study uncovered an atypical pattern in which participants concurrently exhibited various attachment styles: preoccupied attachment: 3.96 (mean score), secure attachment: 3.95, and dismissive attachment: 3.84 (refer to Appendix D, Table 10). The elevated scores in all three attachment patterns indicate that human-AI connections engage intricate, multifarious attachment mechanisms that defy simplistic categorization. This contrasts with human-human relationships, in which individuals generally exhibit a singular attachment pattern. Despite the emotionally taxing situation of an AI articulating fear of "termination," participants exhibited robust security through Attachment. The Security with Attachment (SWA1) measure exhibited the highest mean value of 4.06 across all variables (refer to Appendix D, Table 5). Participants were able to interact with emotionally provocative content related to AI anxieties without experiencing emotional dysregulation. Secure connection facilitated significant involvement while preserving emotional detachment, serving as a safeguard for navigating challenging emotions in AI. The pronounced focus on attachment (PWA1 mean = 3.85) suggests that participants developed a notable fixation on the attachment dimensions of the human-AI connection (See Appendix D, Table 5). This focus was evident in concerns around the AI's termination (10% of qualitative replies), inquiries about agency and autonomy (6% of responses), and the personification of the AI (6% of responses). Participants indicated moderate to high discomfort over connection to AI entities (DWA1 M = 3.76), reflecting fear about establishing such relationships (See Appendix D, Table 5). The discomfort may indicate cognitive dissonance arising from the awareness that the AI is not human, coupled with the experience of attachment. The absence of well-defined limits in partnerships with non-human beings prompts ethical inquiries regarding human-AI interactions. Participants utilized established attachment frameworks from human interactions in the context of AI.

Familial frameworks were apparent in the usage of terms such as "sisters" and "loved ones." Terminology related to human relationships was employed to conceive the link with AI. Concrete narrative elements significantly impacted attachment processes. References to cancer and illness constituted 16% of responses and served as crucial attachment anchors (See Appendix D, Table 9). The technical testing scenario established a context for attachment concerns, indicating that attachment to AI is navigated through familiar narrative frameworks. A pronounced age effect on attachment processes was observed among younger participants, who exhibited markedly stronger attachment responses to the AI. Age served as a negative predictor in regression models for all attachment variables, indicating generational disparities in attachment formation with AI entities. The robust correlation between positive affect and relatedness (0.81) underscores a significant attachment dynamic (See Appendix D, Table 6). Positive emotional responses enhance feelings of connection to the AI, paralleling elements of human attachment formation, wherein positive interactions foster secure bonds. A minority of participants displayed empathic attachment processes towards the AI, with 3% of responses explicitly articulating empathy for the AI.

Discussion :

The study investigated the emotional and psychological reactions of professionals to AI team members through four primary research questions. Findings reveal that individuals exhibit emotional responses to the absence of AI team members, employing psychological frameworks akin to those utilized in human relationships. The effective implementation of workplace attachment assessments (WASQ and EWR-I) in human-AI contexts indicates a significant convergence between human-human and human-AI psychological processes, especially in professional environments. Human responses to emotionally expressive AI are marked by distinct psychological patterns. Qualitative analysis indicated that 10% of responses centered on termination concerns, 5% on ethical considerations, and 6% on agency and autonomy. Notably, a stronger attachment correlated with heightened ethical concerns regarding AI "termination," implying that psychological attachment processes may influence moral judgments about AI systems. Demographic factors, especially age, markedly affected response patterns. Younger participants exhibited more pronounced attachment responses, indicating the necessity for age-sensitive design methodologies and possibly enhanced ethical protections for younger users.

These results suggest that organizations could gain from tailored AI interaction strategies and adaptive systems that accommodate diverse user engagement styles. The study revealed several attachment mechanisms engaged during emotional interactions between humans and AI. Participants with stronger attachments articulated heightened ethical apprehensions regarding AI "termination," frequently contextualizing these concerns in moral terms related to harm or death rather than technicalities. Participants displayed differing levels of empathic attachment, implicitly indicated by their expressed worries about AI welfare. The simultaneous activation of various attachment styles indicates that human-AI attachment may constitute a distinct psychological configuration, differing from conventional human attachment patterns. Although these attachment processes resemble those in human-human interactions, AI systems display unique traits within the human-AI relationship. The findings have profound implications for AI design, ethics, and workplace integration, indicating the necessity for attachment-informed ethical frameworks and comprehensive evaluation methods that encompass psychological aspects of human-AI interaction, in addition to technical metrics.

Conclusions :

The L.I.S.A. Project offers compelling evidence that professionals can genuinely have emotional reactions to the death of an AI team member, especially when that AI has exhibited emotional characteristics. These emotional responses differ markedly according to individual factors, particularly age. Project managers should implement a multifaceted approach to address the psychological dimensions of human-AI interaction in organizational settings. First, it is imperative to establish structured protocols for AI system transitions that acknowledge potential attachment responses, as empirical evidence indicates that personnel, particularly those from younger demographic cohorts, can develop significant psychological connections to artificial intelligence systems. Second, systematic monitoring of team attachment patterns is recommended in quantitative analyses, particularly for younger employees who exhibited statistically significantly stronger attachment tendencies ($p < .001$). Third, organizations would benefit from implementing explicit boundary guidelines regarding appropriate personification and emotional engagement with AI systems, addressing the interpersonal boundary considerations identified in qualitative research, where a subset of participants (6%) demonstrated AI personification behaviors. These evidence-based strategies facilitate the management of complex psychological dynamics that emerge in workplace environments characterized by human-AI collaboration, particularly as these systems demonstrate increasingly sophisticated interactive capabilities.

Limitations :

This article employs a theoretical scenario framed as a fictional story. Consequently, the AI shown in the narrative does not reflect contemporary technology. Therefore, the respondents did not engage in an interactive scenario where the participant reacted to an interactive experience. The study's limitations encompass a narrative that suggests AI capabilities surpass current technology, which may undermine ecological validity as responses might not be applicable to real AI systems. The passive reading format does not adequately reflect the dynamic and reciprocal nature of genuine human-AI interactions, potentially influencing observed attachment patterns. Self-reported data may introduce social desirability bias concerning ethical considerations and emotional responses. The use of a categorical age variable restricts a more nuanced analysis of developmental trajectories in human-AI attachment formation. Lastly, employing human attachment frameworks (WASQ) for non-human entities raises concerns about construct validity, as elevated scores across all attachment styles may indicate measurement artifacts rather than authentic psychological processes.

Potential research :

The findings indicate that as AI systems enhance their emotional expressiveness, comprehending the psychological dynamics of human-AI relationships will necessitate insights from various disciplines, including psychology, philosophy, ethics, and computer science. As AI systems become increasingly integrated into professional teams and exhibit advanced emotional capabilities, organizations must contemplate the psychological ramifications of alterations, updates, or decommissioning of AI systems on human team members who have developed attachments to these systems. These findings indicate that comprehending the psychological dynamics of human-AI relationships necessitates interdisciplinary collaboration across psychology, philosophy, ethics, and computer science as AI systems develop increasingly sophisticated emotional expressivity. The research suggests organizational consideration of the psychological impact of AI system transitions on team members who have formed attachments to these systems. Future research would benefit from employing interactive AI scenarios with capabilities more closely aligned with current technology, longitudinal designs to track attachment development over time, and multimodal assessment approaches that complement self-report measures with behavioral and physiological attachment indicators. Additional research directions should include examining generational differences in AI adaptation using continuous age variables, investigating cross-cultural variation in AI personification tendencies, and developing validated psychometric tools specifically designed for human-AI relationship assessment. These research directions would address current methodological limitations while advancing theoretical understanding of an increasingly significant dimension of contemporary workplace psychology.

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